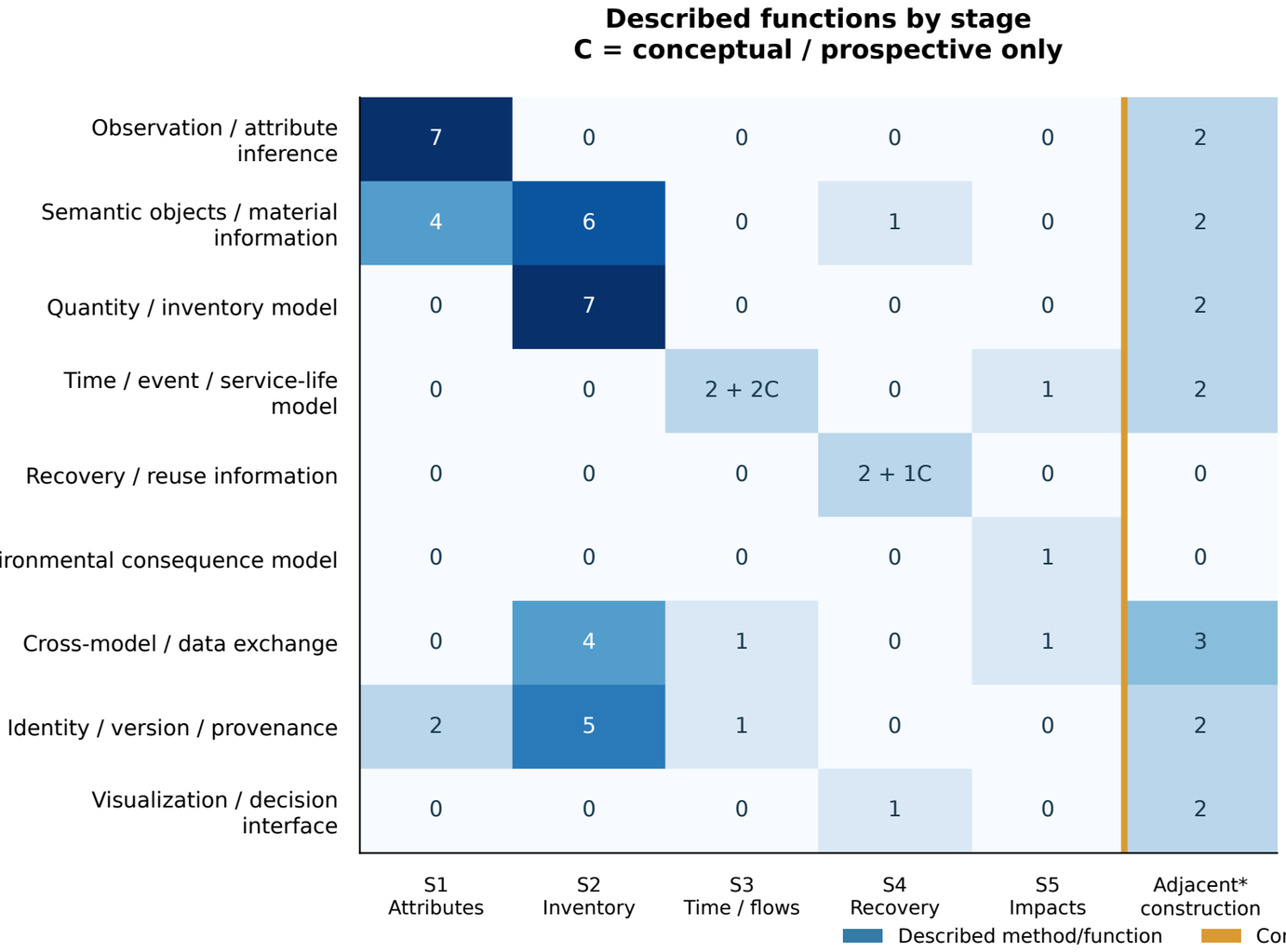
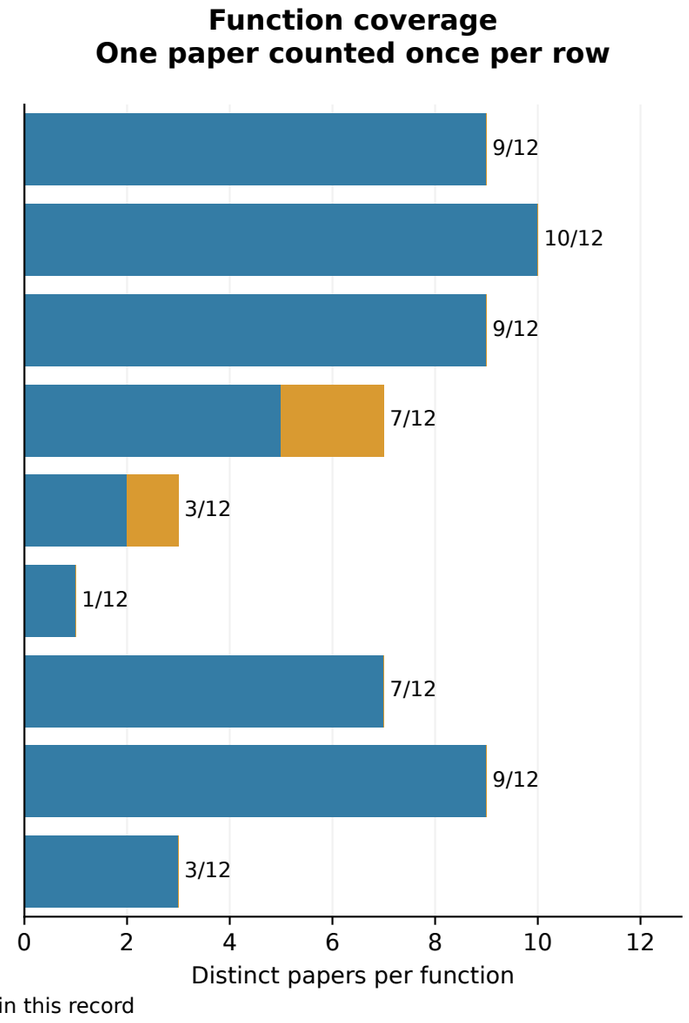
02 Stages × Digital Twin and enabling functions

Twelve assessed domain / enabling studies. Functions are counted; this is not a count of twelve validated DTs.





Stage records with described functions: S1=7, S2=7, S3=2, S4=2, S5=1, X=3. Stage memberships overlap.

* Adjacent construction logistics, cost and schedule studies are not relabelled as building-stock turnover.

S3 includes retrospective event detection; S4 includes reuse information; neither label proves probabilistic flow/recovery modelling.

Zero = not positively coded, not proven absence. Sparse S4/S5 coverage is a property of this subset, not the entire literature.